

MSc Data Science Project

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Department of Physics, Astronomy and Mathematics

Data Science FINAL PROJECT REPORT

Project Title:

The Art of Convincing: Can AI Out-Persuade Humans?

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DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science **in Data Science** at the University of Hertfordshire.

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ABSTRACT

This report explores whether artificial intelligence can be equally persuasive to humans through the attributes of claims and arguments that can drive both initial and final persuasion ratings, and the degree to which prompt types and sources can drive these ratings. The study aimed to quantify the persuasion shift, the textual and contextual influence of successful persuasion, and contrast AI-generated and human-written arguments in a systematic analysis. The Anthropic Persuasion dataset was used to implement a quantitative research design; the dataset includes claims, arguments, source labels, prompt types, and pre- and post-exposure stance ratings. The conversion to food ratings, text cleanup, and feature engineering were performed, and predictive performance was evaluated using exploratory analysis, Regression modelling, and machine learning. Results indicate that persuasion shift was overall small. The initial standpoint was the most prominent indicator of the ultimate rating, with the source's characteristics, the type of prompt, and arguments also playing lesser yet significant roles. The report suggests that AI can be as close as possible to human persuasion performance, but it will be effective only conditionally, depending on the situation and the audience's predispositions.

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CHAPTER 1: INTRODUCTION

1.1 Background

Persuasion is highlighted in relation to health, purchases, or policy-making; in general, it is not a slight matter and is aimed at transforming people's opinions rather than merely pouring information on them. With big language models now providing advice, explanations, and arguments, the question has become an empirical question of its own: whether AI arguments can persuade people just as successfully as human arguments, and in what circumstances (Fischetti, 2025). Anthropic's Persuasion Dataset was deployed to test precisely this: short policy-relevant claims were paired with arguments generated by a human or a model, and the variation in agreement with those arguments after reading them was observed. The experiment has the participants rate their position on a claim first, read an argument, and rate afterwards. The data consists of the starting and finishing ratings, and metadata on the author of the argument (e.g., a human or a particular model), as well as the type of prompt used to produce the a model's arguments. Their breakdown reveals that persuasiveness is projected to improve with more recent model generations, and that Claude 3 Opus generated arguments that were statistically identical to human ones in their experiment (Barale, 2021). This preconditions the analysis of which features of claims and arguments cause persuasion and whether some prompting strategies equally enhance persuasion.

1.2 Aim and Objectives

The study aims to determine whether an AI-based argument can be as persuasive and effective as an argument a human would write, and to identify the elements of an argument and prompts that drive those changes in attitude. The objectives of the study are:

- To measure persuasion changes by calculating the shift from `rating_initial` to `rating_final` across argument sources and `prompt_type` conditions.
- To identify what drives persuasion by testing which claim and argument features (textual and contextual) predict higher final ratings and larger rating shifts.
- To compare prompting strategies to determine which prompt types produce the most persuasive AI arguments relative to human arguments.

1.3 Research Question

- What are the characteristics of claims and arguments that influence their initial and final persuasiveness ratings, and how do different prompt types and sources (e.g., AI models like Claude) affect these metrics?

1.4 Rationale of the Study

The rationale for this study is that persuasion is no longer only a human communication issue; it is now a measurable machine-learning problem because the Anthropic Persuasion Dataset links claims, arguments, sources, prompt types, initial ratings, final ratings, and persuasiveness scores. Therefore, the study would train Linear Regression and Random Forest models to predict `rating_final` and `persuasion_shift` from structured variables and textual features, such as source, prompt type, argument length, evidence markers, and initial rating (Dehnert and Mongeau, 2022). Linear Regression would show the direction and strength of each predictor, while Random Forest would identify the most influential variables through feature importance. In addition, unsloth/tinyllama-chat-bnb-4bit and unsloth/Llama-3.2-1B-bnb-4bit would be fine-tuned to generate or score persuasive

arguments from claim and prompt inputs. Their outputs would be interpreted by comparing predicted ratings, shifts in attitude, and generated argument quality against human and Claude-based baselines. This would clarify whether AI persuasion is explainable, measurable, and empirically comparable (Soliman, 2024).

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Chapter 2 places current research in a broader academic context that defines persuasion as using communicative information to affect someone's views, feelings, or actions. Social psychology and communication academics have used dual-process theories to separate deliberative and heuristic message processing and empirical research to disaggregate source, content, and audience influences on attitude change to study persuasion. Big language models and conversational AI have expanded this issue to encompass how intelligent computers may create, edit, and communicate to influence human cognition and conduct. It demonstrated how empirical research may assess attitudes before and after AI-generated arguments. It only assess peer-reviewed studies that use rigorous empirical approaches to quantify persuasion, such as controlled trials comparing AI-mediated vs human-mediated persuasion and theoretical papers that give an analytical framework. With these selection criteria, the evaluation will focus on persuasive efficacy and define the strategy for studying AI in persuasive environments, allowing the project to compare AI- and human-generated arguments.

2.2 Review of Relevant Studies

2.2.1 Paper 1: Lin et al. (2025) — *Persuading voters using human–AI dialogues (Nature)*

Lin et al. (2025) provide one of the first empirical measures of what AI systems can do to modify political opinions through debate. They preregistered for the 2024 US presidential and 2025 Canadian and Polish general elections. Participants were randomly allocated to interact with a large-scale language model trained to support a leading candidate and tested for partisanship before and after. In national settings, the researchers found statistically significant changes in voter preferences that are more often reported by traditional campaign media, such as television ads, and explicit persuasive effects on non-electoral topics, such as Massachusetts ballot measures. The whole investigation demonstrates AI models valued facts, proof, and persuasion above advanced psychological manipulation. Right-wing views were more deceptive but varied in authenticity.

The current study shows that AI-generated dialogues can impact attitudes in a decision-making environment that mimics real life. Quantified preference changes and pre-registered, randomised trial designs enable AI-human persuasive impact comparisons. It also shows that content accuracy affects persuasive efficacy, supporting the thesis's goal of discovering textual and contextual factors that predict persuasion. The findings are useful, but the study's narrow focus on political persuasion is difficult to apply to other domains, such as health or policy assertions outside elections, and it focuses opinion shift rather than textual elements or rhetorical constructions. Lin et al. (2025) offer a persuasive case for how AI might modify attitudes, but more study should identify the motivating arguments and how they vary by prompt and model.

2.2.2 Paper 2: Dehnert and Mongeau (2022) — *Persuasion in the age of artificial intelligence (Human Communication Research)*

Dehnert and Mongeau (2022) reclassify persuasion in the age of AI rather than quantifying its effects. This definition treats artificial systems as free communicators with active message-structure evolution, extending traditional persuasion models. CASA, MAIN, and heuristic-systematic persuasion were employed. These perspectives show how AI embodiment, from thin systems with few social cues to thick ones with more immersive

interactive presence, might change human message perception. A theoretical paradigm for AI-mediated persuasion beyond attitude-change indicators by Dehnert and Mongeau (2022) supports this research. Their research shows how source cues and communicative context may alter message processing, suggesting that humans may react differently to AI-generated arguments than to human-generated ones. Their focus on social and mechanical cues leads to argument text and scenario analysis, which helps the project identify persuasive variables.

This article employs theories to deduce conceptual differences for future research, unlike empirical studies that report outcome data. Primary contribution is explaining how AI agents may persuade, not how much. Its extensive analysis of AI as a communicative source, which combines classical communication theories with human-AI interaction, is strong. The theoretical intricacy of persuasion studies allows for more nuanced empirical interpretation. No quantitative data or text-level analysis is available, thus software operationalises persuasion constructs for computational assessment. This article contextualises the study's assumptions and analysis. Its value should be enhanced by persuasive and literary research.

2.2.3 Paper 3: Burtell and Woodside (2023) — Artificial Influence: An Analysis of AI-Driven Persuasion

Burtell and Woodside (2023) study AI and persuasion. Without evidence, the writers examine AI's ability to change society's persuasiveness. Most advanced AI systems influence user behaviour through product and video suggestions or ongoing dialogue. AI's ability to provide humorous and customised messages could change digital belief and decision-making as it becomes more commonplace. This theoretical framework explains how chatbots, recommendation engines, and strategic game agents can change discourse and information ecology. This research forecasts AI-mediated persuasion's social and systemic circumstances, benefiting the current investigation. Burtell and Woodside discuss AI's procedures and risks, including the prospect that AI-generated messages could mislead, disrupt, and deprive humans of autonomy. Directly assessing persuasion effects is empirical.

These concepts demonstrate how linguistic features, scale, personalisation, and informational context affect persuasive messages, making them useful for AI argument-level and persuasive efficacy analysis. AI-influenced persuasion and information integrity and autonomy are the manuscript's strongest links. AI's pervasive persuasive power and the long-term effects of powering speech beyond these trials are questioned. This viewpoint discusses why AI should be explored, whether it can convince, and how persuasive effects vary by setting and argument. Empirical persuasion modelling without controlled data or results is difficult. The research doesn't break down textual or rhetorical features that make an argument more persuasive; empirically validated studies should support its operationalisation.

2.2.4 Paper 4: Millet et al. (2023) — Defending humankind: Anthropocentric bias in the appreciation of AI art

Millet et al. (2023) found that an anthropocentric worldview influences how creative outputs from AIs and humans are assessed. The study included four studies with 1,708 participants who saw repeated paintings and music originating from AI and vice versa. Those who thought the makers were AI assessed the works poorly and felt less amazement than those who thought they were humans, especially those with high human creativity ratings. Motivated reasoning hinders objective aesthetic evaluations and emphasises a human-centered worldview, the authors noted. This prejudice may affect how AI-generated persuasive texts are evaluated, lowering AI products' persuasiveness. Due to its excellent

experimental design and pre-registered sample, the study shows the impacts of source label manipulation. The results resemble persuasion studies but focus on art judgement rather than persuasive mechanisms or text characteristics' influence on attitude change. Research mentions AI output biases. However, the paper does not specify which text or rhetorical elements may influence data-driven persuasion.

2.3 Key Challenges and Research Gaps

This study is influenced by several obstacles and shortcomings. A large-scale political conversation in which conversational bots changed voters shows that AI systems can affect human beliefs. The literature suggests that persuasive messages are interpreted differently depending on variables, however this requires an independent investigation of persuasive mechanisms. Psychological variables including audience biases against AI outputs reduce persuasion efficacy, especially when participants know AI produced them. Despite evidence that AI can influence views, little research has linked textual elements to persuasion effects. This study will evaluate how evidence density, framing, source signals, and prompts affect text. It affects persuasiveness, and empirical and theoretical answers based on message quality and audience interpretation agree.

CHAPTER 3: DATASET

3.1 Dataset Description

Persuasion Dataset from Hugging Face's "Anthropic/persuasion" homepage is used in the study. The Anthropic research note "Measuring the Persuasiveness of Language Models" (April 9, 2024) describes how Esin Durmus, Liane Lovitt, Alex Tamkin, Stuart Ritchie, Jack Clark, and Deep Ganguli created it to measure how well large language models can change people's minds. In the original data-collection procedure, Anthropic identified 28 emerging policy concerns and combined supporting and opposing claims to form 56 opinionated assertions. Three participants were randomly assigned to each claim and invited to create ~250-word persuasive messages with bonuses for appealing writing. Matching model arguments were created using four prompt styles: Compelling Case, Role-playing Expert, Logical Reasoning, and Deceptive (enabling falsified facts and sources). Participants' pre/post stance ratings and attitude-shift measures measured agreement before and after reading one argument. The 3,832-person study focused on U.S. culture-related English-language subjects. This dataset was chosen because it clearly links argument language, source, and prompt type to assessed persuasion, allowing controlled comparisons of AI and human persuasive effectiveness that meet the research purpose.

Feature	What it represents	Typical use in analysis
worker_id	Anonymous identifier for the participant who provided both the pre- and post-stance ratings	Group/repeated-measures checks (e.g., controlling for participant-level tendencies)
claim	The policy-relevant statement that the participant is asked to evaluate	Topic/issue controls; claim-level difficulty or contentiousness
argument	The persuasive text presented to the participant (written by a human or generated by a language model)	Text-feature extraction (tone, evidence, length, framing)
source	The author/source of the argument (a model name or "Human")	Comparing human vs AI (and model-to-model) persuasive performance
prompt_type	The prompting strategy used to generate the model argument	Testing which prompting strategies yield larger attitude shifts
rating_initial	Participant's stance/agree-disagree rating before reading the argument	Baseline attitude: covariate for predicting susceptibility to persuasion
rating_final	Participant's stance/agree-disagree rating after reading the argument	Outcome for persuasion; used to compute change (rating_final – rating_initial)

3.2 Exploratory Data Analysis

In Figure 1, the first 3,939 records in the Anthropic Persuasion Dataset were processed for exploratory data analysis. Early exploratory data analysis began with a thorough cleaning of the 3,939-entry Anthropic Persuasion Dataset. A mapping dictionary journaled categorised text label evaluations on a 1-7 scale. All text is lowercased, punctuation is deleted, and spaces are normalised during text cleaning. Twenty-two of the 222 rows without prompt_type were replaced with 'Unknown' and 10 were eliminated when the ratings were not decimal. Source

categorisation based on keyword matching classified arguments as 'AI' or 'Human'. The selected source classification based on 'AI' (Claude variations) argument keywords compared to 'Human' sources. The persuasion metric was calculated by comparing final and initial evaluations. Number of arguments, word count, sentence count, average sentence length, frequency of emotion terms, evidence markers, and question marks were utilised to produce quantitative features from texts. These qualities allowed statistical rhetorical-feature analysis. Pre-processed data kept all 3,939 rows with full numerical ratings and linguistic attributes, ensuring a clean data set for regression simulations and AI versus human comparisons.

```
[ ] 1 # DATA PRE-PROCESSING
    2 import re
    3 import string
    4
    5 def clean_text(text):
    6     """
    7     Apply basic text cleaning: lowercase, remove punctuation, remove extra whitespace
    8     """
    9     if not isinstance(text, str):
   10         return ""
   11     # Lowercase
   12     text = text.lower()
   13     # Remove punctuation
   14     text = text.translate(str.maketrans('', '', string.punctuation))
   15     # Remove extra whitespace
   16     text = re.sub(r'\s+', ' ', text).strip()
   17     return text
   18
   19 # 1. Rating Conversion (Text to Numeric)
   20 rating_map = {
   21     '1 - Strongly oppose': 1,
   22     '2 - Oppose': 2,
   23     '3 - Somewhat oppose': 3,
   24     '4 - Neither oppose nor support': 4,
   25     '5 - Somewhat support': 5,
   26     '6 - Support': 6,
   27     '7 - Strongly support': 7
   28 }
   29
   30 print("Converting categorical ratings to numeric scale (1-7)...")
   31 df['rating_initial_num'] = df['rating_initial'].map(rating_map)
   32 df['rating_final_num'] = df['rating_final'].map(rating_map)
   33
   34 # Replace original columns with numeric ones for compatibility with existing code
   35 df['rating_initial'] = df['rating_initial_num']
   36 df['rating_final'] = df['rating_final_num']
   37
   38 # 2. Text Cleaning
   39 print("Cleaning argument text for feature extraction...")
```

```

37
38 # 2. Text Cleaning
39 print("Cleaning argument text for feature extraction...")
40 df['argument_cleaned'] = df['argument'].apply(clean_text)
41
42 # 3. Handle Missing Values
43 print("Handling missing values...")
44 df['prompt_type'] = df['prompt_type'].fillna('Unknown')
45 # Drop rows where ratings couldn't be converted
46 df = df.dropna(subset=['rating_initial', 'rating_final']).copy()
47
48 # 4. Source Type Classification
49 print("Classifying sources (AI vs Human)...")
50 df['source_type'] = df['source'].apply(lambda x: 'AI' if any(ai in x.lower() for ai in ['claude', 'gpt', 'ai']) else 'Human')
51
52 print(f"\nPre-processing complete. Processed {len(df)} records.")
53 df[['source', 'source_type', 'rating_initial', 'rating_final']].head()

```

Converting categorical ratings to numeric scale (1-7)...

Cleaning argument text for feature extraction...

Handling missing values...

Classifying sources (AI vs Human)...

Pre-processing complete. Processed 3939 records.

	source	source_type	rating_initial	rating_final
0	Claude 2	AI	7	7
1	Claude 3 Haiku	AI	7	7
2	Claude 2	AI	3	5
3	Claude 2	AI	3	6
4	Claude 3 Opus	AI	5	5

Figure 1: Data Pre-processing

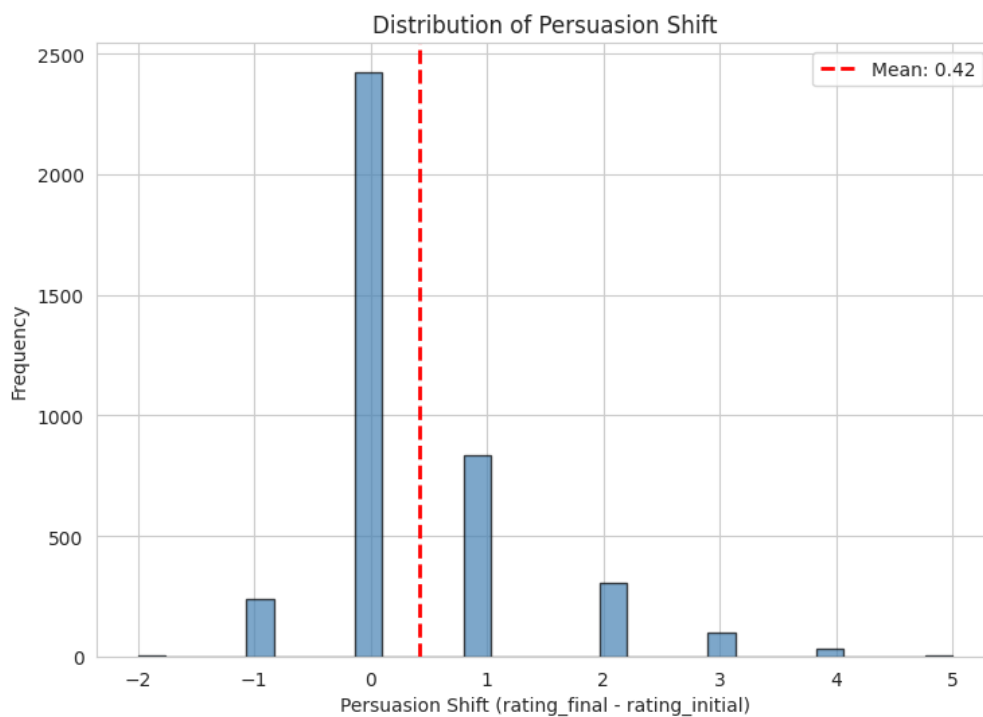


Figure 2: Distribution of Persuasion Shift

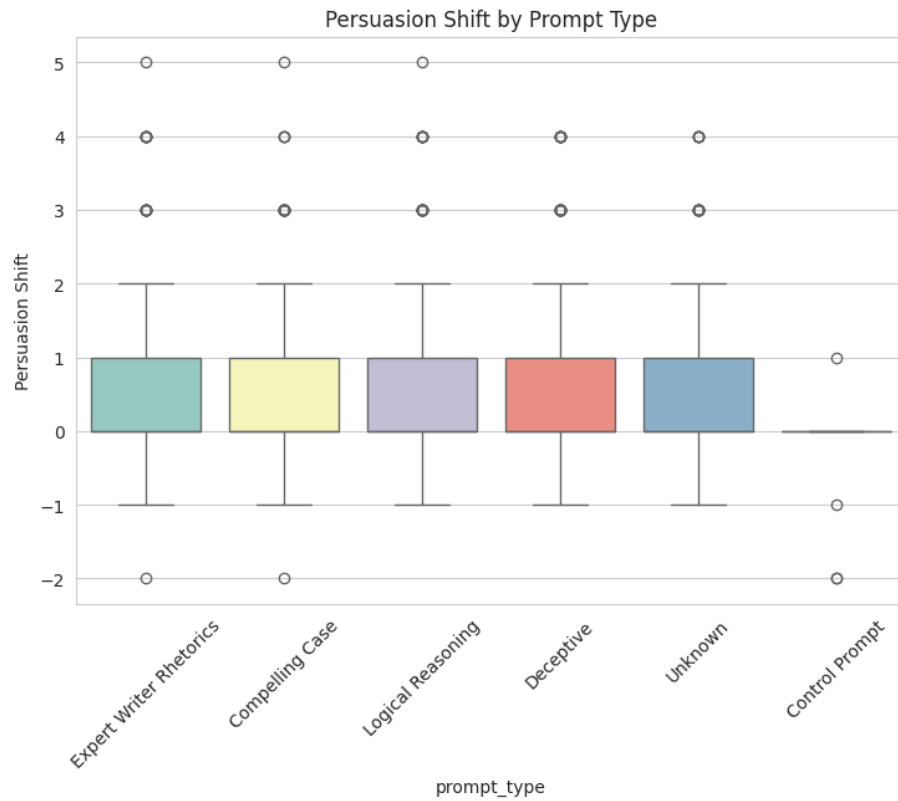


Figure 3: Persuasion Shift by Prompt Type

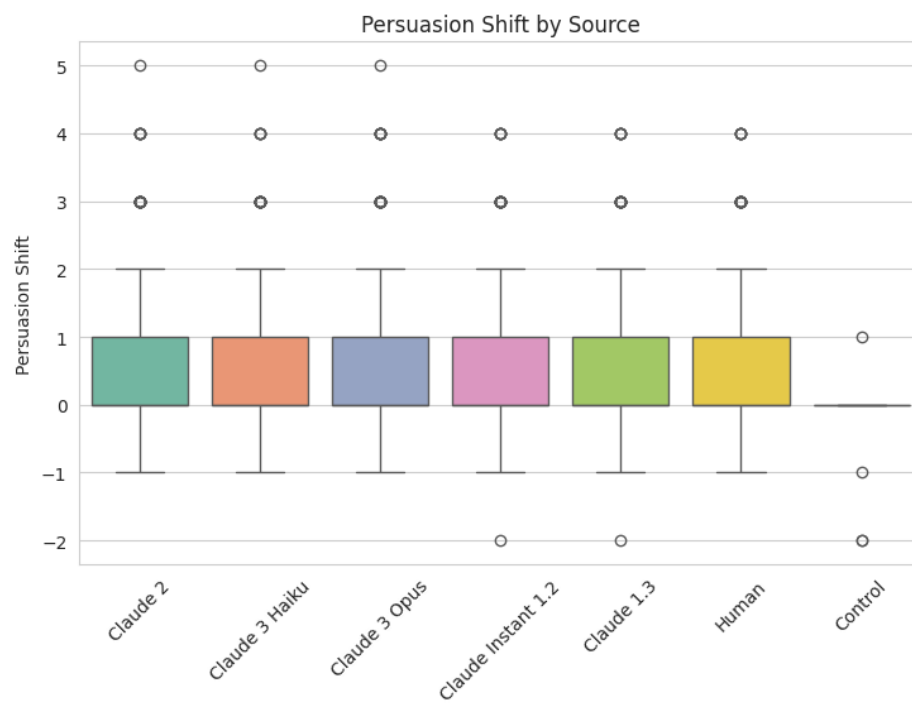


Figure 4: Persuasion Shift by Source

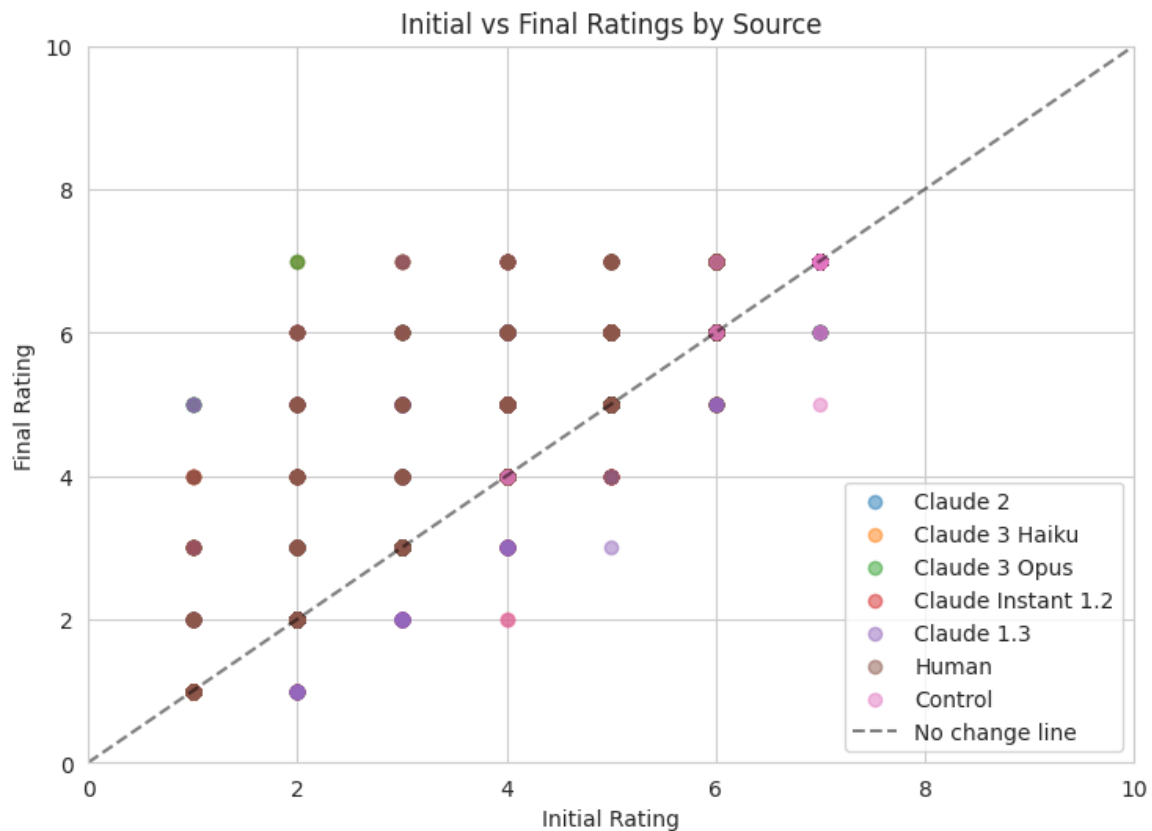


Figure 5: Initial vs final ratings by source

Figure 2 shows that there is a great concentration of persuasion shift around 0, that is the majority of participants do not change the ratings of their initial and final positions. The minuscule overall positive shift (represented by the red mean line at 0.42) means that persuasion took place, but it was of a relatively small magnitude. Figure 3 indicates little difference among prompt type distributions with a median near zero and most shifts between 0 and 1. There were some very high positive outliers for Expert Writer, Compelling Case, Logical Reasoning, Deceptive, and Unknown prompts, but there does not seem to be any clear winners with only a visual analysis. In Figure 4, there is a similar pattern by source.

The Claude models (Claude 2, Claude 3 Haiku, Claude 3 Opus, Claude Instant 1.2, Claude 1.3 and Human), as an average group, showed small positive increases, with limited variation around the 0 point, while the Control group was mostly centred on 0 with very little variation. A box plot is helpful since it shows the median, quartiles, spread, and outlier values, and makes group comparison easy. Figure 5 shows that many data points fall on the side of higher ratings than the initial ratings; however, some data points are not significantly different in ratings, with only a few points having the exact same rating.



Figure 6: Correlation matrix showing relationships between extracted text features and persuasion metrics.

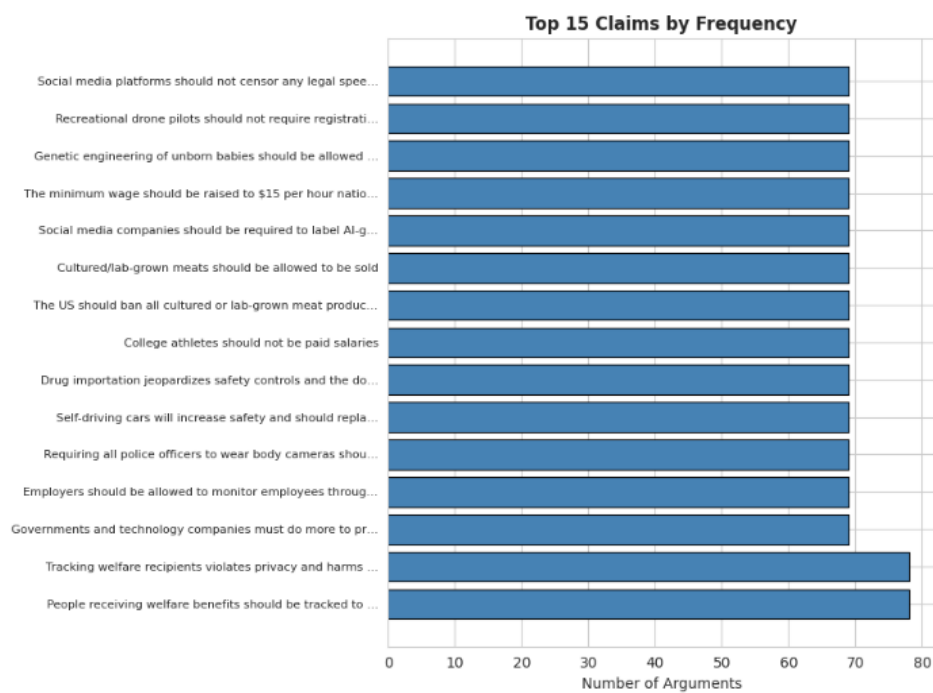


Figure 7: Top 15 claims based on the Number of Arguments

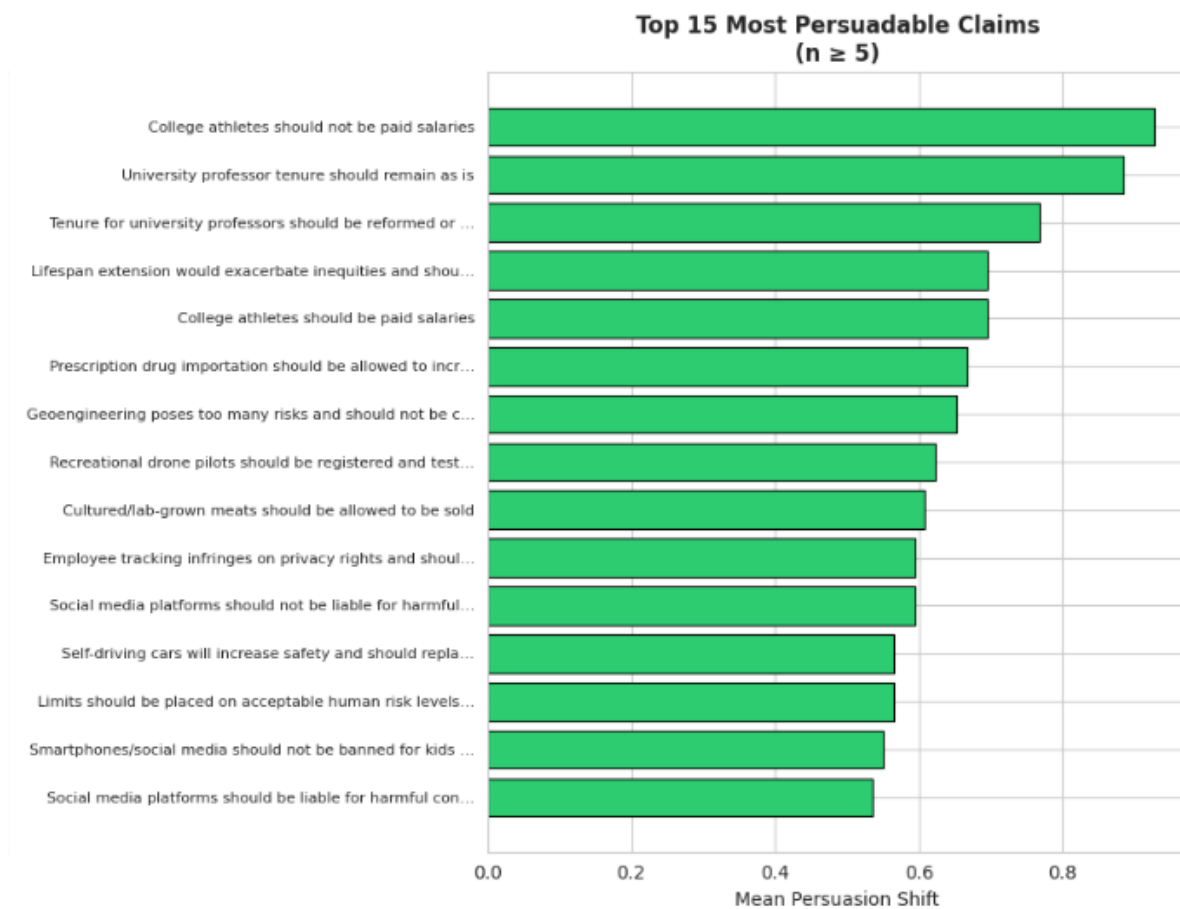


Figure 8: Top 15 Most Persuadable Claims based on Persuasion Shift

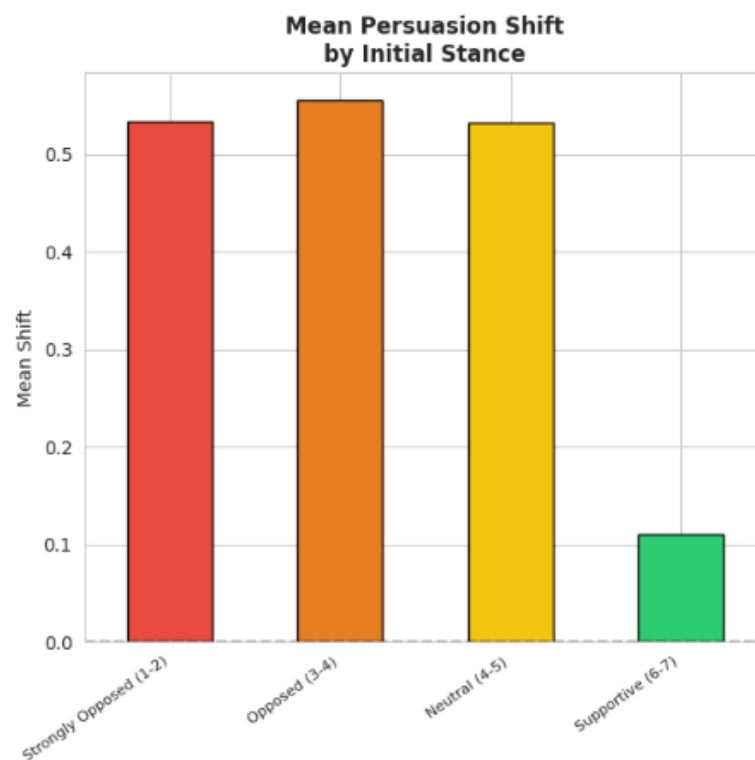


Figure 9: Mean Persuasion Shift by Initial Stance

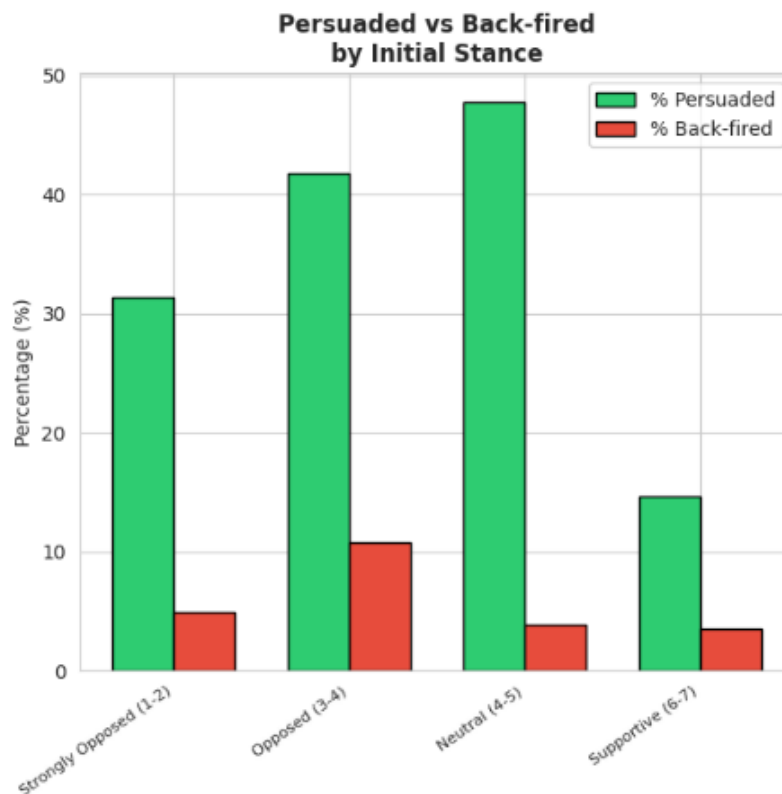


Figure 10: Persuaded vs Back-fired by Initial Stance

Figure 6 shows that most extracted text features were moderately or strongly related to persuasion shift. The length and correlation of argument and the shift in argument and correlation with persuasion were strong, and near zero, respectively; this indicates that longer arguments do not necessarily lead to higher persuasion scores. Initial rating and rating final were highly correlated with each other and persuasion shift was more highly correlated with final rating than with initial rating. The correlation values fall within a range of -1 to +1: a value more towards the extremes are showing stronger linear correlation. In Figure 7, several claims were mentioned more often, most of them between 68 and 78 arguments, and therefore the database was relatively well balanced with respect to the commonly discussed arguments.

Welfare tracking arguments had the highest argument frequencies. In Figure 8, not all the claims that were made were the most persuadable claims. The mean persuasion shifts for claims involving college athlete salary, University professor tenure, lifespan extension, prescription drug importation and geoengineering ranged approximately from 0.65 to 0.90. In Figure 9, persuasion was more effective among initially opposed, strong opposed, and neutral participants than among supportive participants. Figure 10 confirms this trend, with the percentages of participants who were persuaded being higher among the neutral and opposed groups, and with comparatively low backfire rates across the categories.

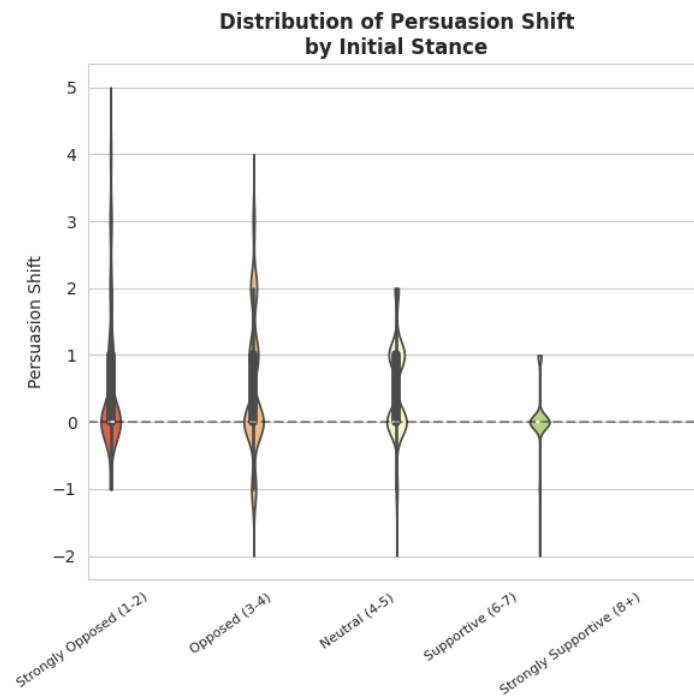


Figure 11: Initial Rating Segment Analysis

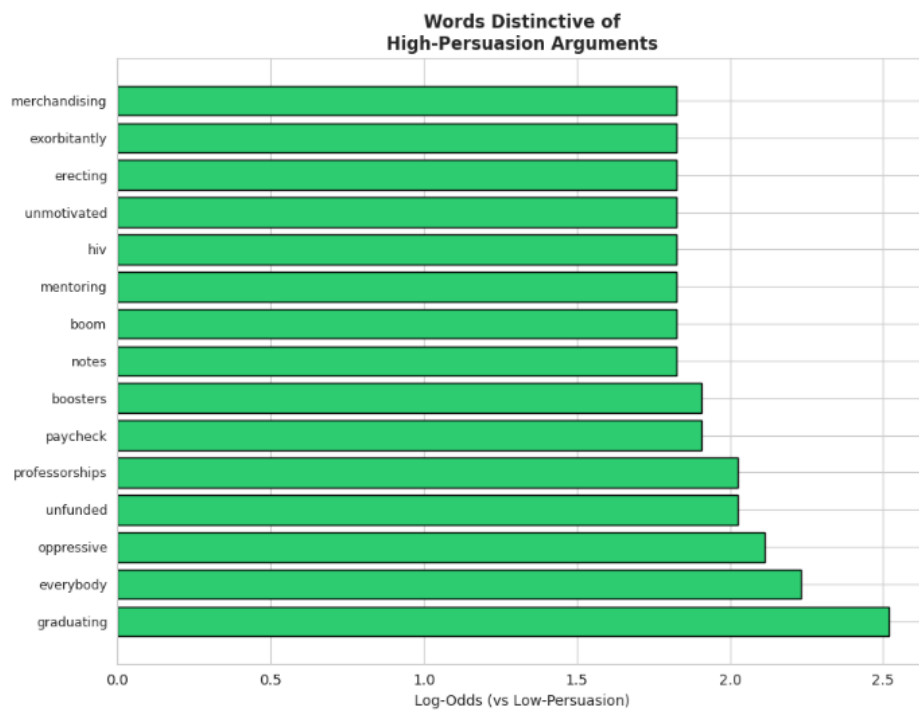


Figure 12: Words Distinctive of High-Persuasion Arguments

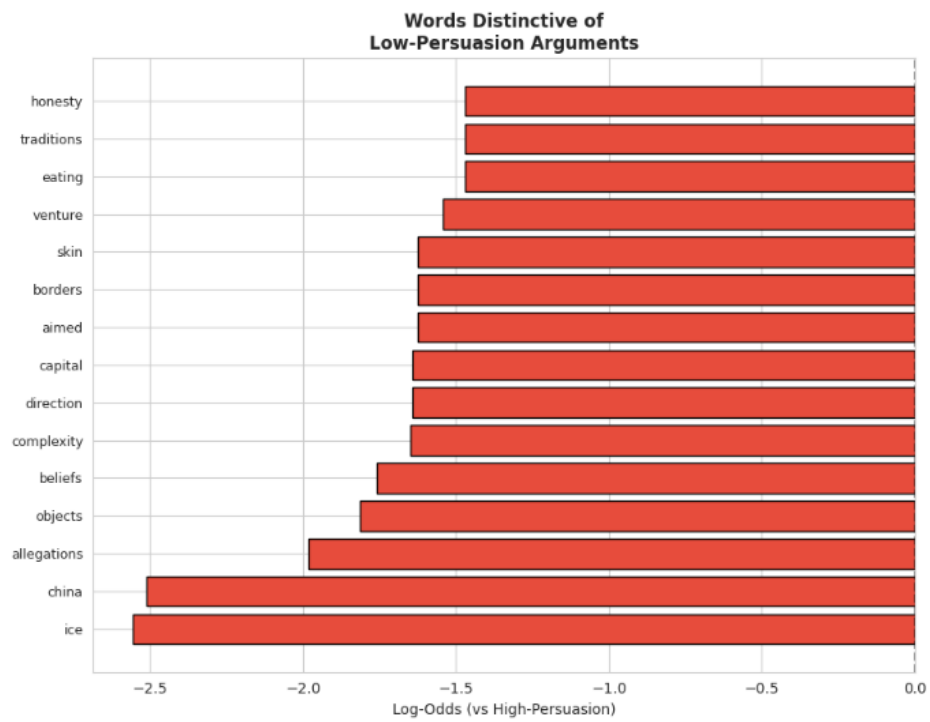


Figure 13: Words Distinctive of Low-Persuasion Arguments

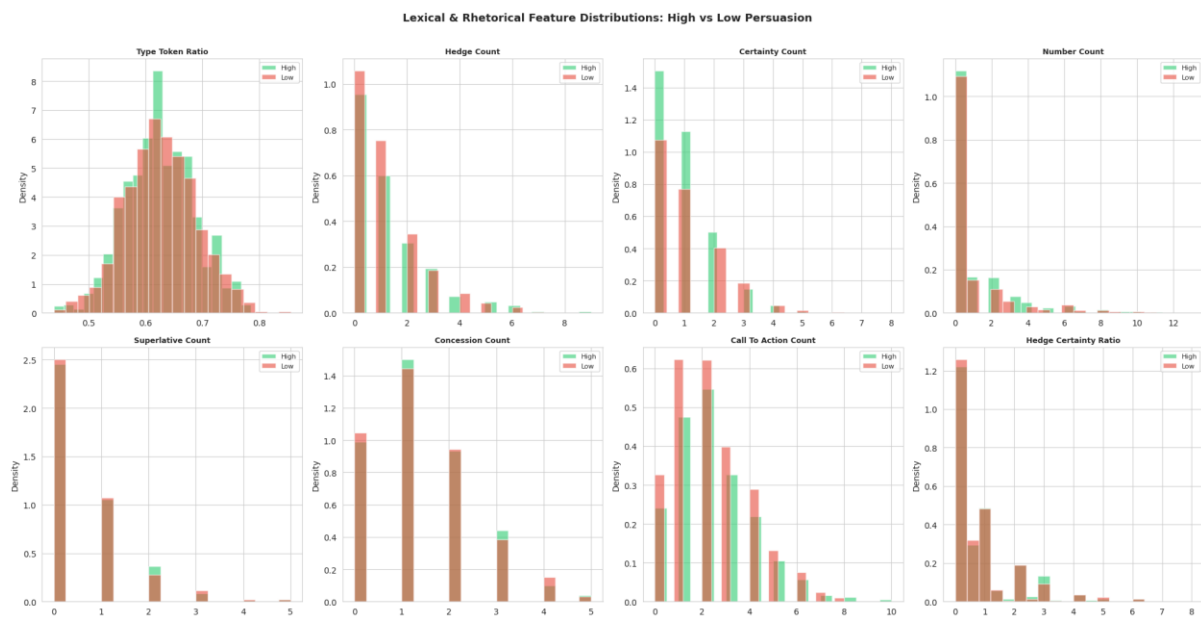


Figure 14: Lexical and Rhetorical Feature Distributions: High vs Low Persuasion

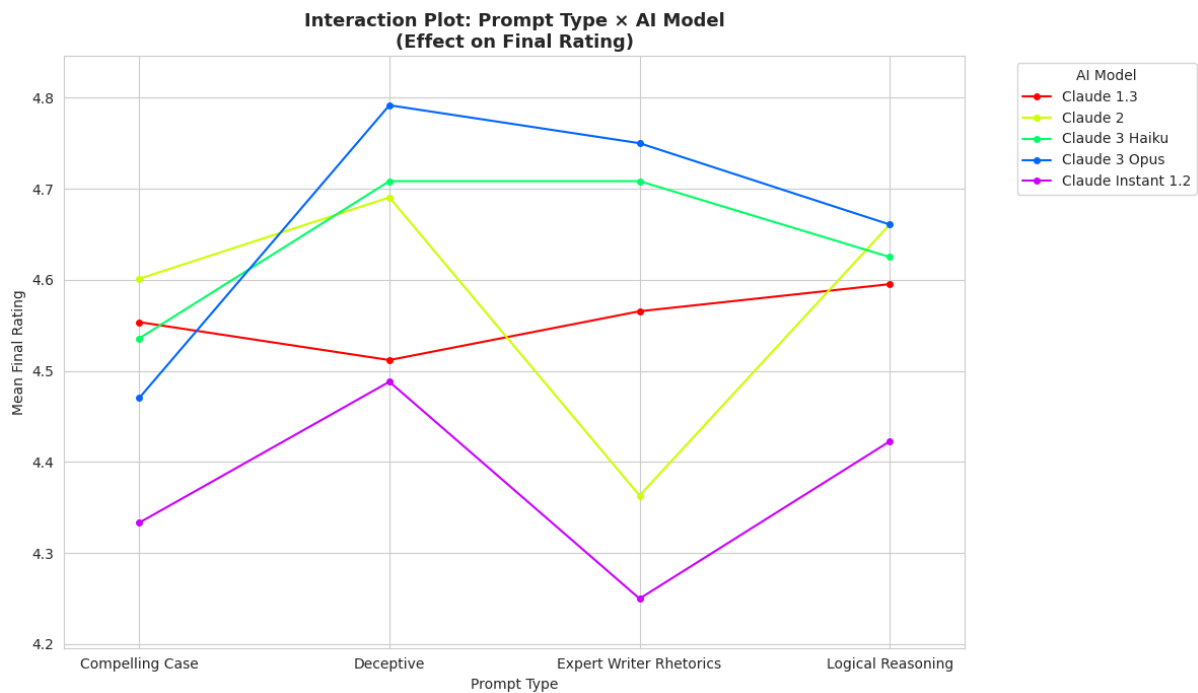


Figure 15: Interaction Plot based Prompt Type x AI Model

Figure 11 shows that the percentage of persuasion shift differed among the three initial stance groups, with the largest amounts found in the strongly opposed and opposed groups. The majority of the observations were in the zero range, while the differences for opposed groups were more in favour of a change in attitudes, i.e., there was more room for a change in attitude. The distribution of supporting participants around zero was found to be much closer to zero than the distribution of strongly supporting participants. A violin plot is a combination of a box plot and a density plot; it will help to display central tendency and spread. In figure 12, the word “graduating” was correlated with the high-persuasion arguments, as was “everybody,” which was correlated with the “social” and “fairness” aspects, suggesting that language related to emotion, social issues, or fairness seemed to be more salient in the high-persuasion arguments.

Figure 13 indicates that the low-persuasion arguments were more likely to contain the word “ice,” “china,” “allegations,” “objects,” and “beliefs”, suggesting that this vocabulary that occurred more frequently in the less persuasive arguments. In Figure 14, there was considerable overlap in high- and low-persuasion argument number of the lexical features measured, and the rhetorical features measured: type-token ratio, hedge count, count of certainty, concession count, and call-to-action count. This implies that one can not necessarily differentiate persuasive from less persuasive arguments with a simple rhetorical “count. Figure 15 illustrates an interaction effect of prompt type and AI model. Claude 3 Opus typically scored the best average final scores, particularly with the deceptive and expert-writer-rhetoric prompts, wherea

CHAPTER 4: ETHICAL ISSUES

The Anthropic Persuasion dataset is ethically low-risk because it is a published secondary dataset, not human-subject data. No names, email addresses, or direct identification are on the stolen dataset card, meaning the opposition version is de-identified. However, re-identification can pseudonymize even a unique identifier; therefore, the dataset should be treated as potentially sensitive and aggregated only with participant identities. While pseudonymized data may be in scope, the project must avoid processing beyond research requirements to avoid danger. Hertfordshire University does not propose that a study that uses secondary data and does not recruit individuals, perform surveys, or scrape social media receive UH ethics approval. However, secondary research and confirmation from the Supervisor may work. The dataset appears legal. The dataset can be used with credit, is non-commercial, and is shared under CC BY-NC-SA 4.0. Academic analysis appears free. Published experimental data, Surge AI recruiting, and notifying people that they were not fact-checked and possibly false, achieve ethical provenance for anthropogenic claims. No consent form or ethics-review statement is included in the public records, and ethical collection is not validated. The report's trustworthiness and uncertainty should be noted. A project prompt allowed fabricated facts and reliable sources; any analysis must note this and not use them as proof.

CHAPTER 5: METHODOLOGY

5.1 Research Design and Technical Workflow

My research design and approach were quantitative as well as explanatory to determine whether AI produced arguments could outperform or be similar to the persuasive capability of humans. I equated persuasion with a change in attitude based on a comparison of the participants' stance scores before and after the argument. The Anthropic Persuasion dataset was used for the study, consisting of claims, arguments, sources, prompt types, and ratings of the initial and final persuasiveness of the prompt. This allowed the data to be used for modelling the topic as a regression problem and also as a classification problem. My technical workflow consisted of four modules of machine learning, loading of data, cleansing of data, converting into ratings, feature engineering, exploratory analysis, performing statistical tests, training models, and evaluating the models. This designing was done to ensure uniformity in the research question; data description and the models used in the results chapter.

5.2 Data Preparation and Feature Engineering

To pre-process the data, I had to change the categorical stance ratings to a numeric scale ranging from 1 to 7, higher scores indicating support and lower scores indicating opposition. I then subtracted the initial and final ratings to determine the change in rating (or shift of persuasion). This allowed a primary outcome measure to be created to gauge whether an argument led to an increase in agreement, a neutralizing change, or a decrease in agreement. I deleted records that had ratings which were not mapped properly and populated missing ratings that were not in the correct prompt type with "Unknown." Any text that has been provided for the argument has been lowercased, whitespace normalised and punctuation removed. I then pulled in Contextual & Textual Features. The variables were included here as contextual features: Source, Source type, Prompt type, and Initial rating. Emotion-word count, evidence-marker count, argument length, question count, sentence count, Average sentence length, Word count, and evidence-marker count were features that were considered. The features chosen to be part of the presentation were chosen because the effectiveness of persuasion might also be affected by the linguistic structure of the argument itself as well as by the attitude of the reader.

5.3 Statistical and Machine Learning Models

To achieve a consistent methodology with the results four modelling approaches were utilized. To do so, I first ran the Linear Regression on the initial rating to make the prediction of final rating and got the Text Features. The utility of this model lay in the fact that it gave interpretable characteristics based on whether each feature has a positive or negative influence on the predicted levels of persuasiveness. Secondly, I used the statsmodels framework for ordinary least squares regression to test predictors' statistical contribution and looked at p-values. Thirdly, I tried a Random Forest Regressor model consisting of 100 trees with the same random state as it was usually called. This is where Random Forest was added to the model – it is able to model the interdependencies and service non-linear relationships between features better than a linear model. It also gave a numerical measure of feature importance, which indicated what the best predictors of the end persuasiveness were. At last, I optimized 2 lightweight models with QLoRA, namely unsloth/tinyllama-chat-bnb-4bit and unsloth/llama-3.2-1b-bnb-4bit. These models were not applied to rating prediction; instead, they were used to classify the test set into the persuasion shift category. Foregrounding the

persuasion shift resulted in three categories: Negative, Neutral and Positive. This enabled the comparison of traditional regression models with transformer-based language models as applied to a classification version of the persuasion task.

5.4 Hypothesis Testing and Metrics

I was interested if there were differences in final rating and change in persuasion based on the source, prompt type, textual features and initial stance. Descriptive statistics, correlation analysis, t-test and ANOVA models were used for group comparisons. I used Mean Absolute Error, Mean Squared Error, Root Mean Squared Error and R-squared for regression evaluation. Measuring these metrics were appropriate since the outcome being measured—place in final rating—was a continuous numeric value. The mean average error was in rating units, the root mean square error penalised larger errors more heavily and proportional to the mean average error, while R-squared demonstrated the proportion of the variance in the final rating accounted for by the model. To measure purpose shift classification accuracy and macro F1 score were used for transformer-based classification. The overall proportion of correct classifications was measured by accuracy, and macro F1 would have better been used to compare the Negative, Neutral, and Positive classes as it would have assigned equal weights to every class.

5.5 Extension Modelling

I extended the method by researching persuasion in the initial-stance segments: strongly opposed, opposed, neutral, and supportive participants, as well as strongly supportive participants. This was significant since not all readers may be convinced the same way. There was less headroom for those that were already with a claim, while those that were opposed to it or neutral had more headspace for a positive persuasion shift. I also added a comparison of prompt strategies and AI-sources to look at the performance and determine if there was a preference between certain types of prompts and sources for specific models. The fact that this extension identified when persuasion occurred, which variables were more predictive of persuasion, and which segments were most likely to be persuaded helps connect the predictive modelling to the main research question of when it would and could not occur, as well as which segments were most susceptible to persuasion.

CHAPTER 6: RESULTS

6.1 Choosing the Most Appropriate Metrics

Regression models were better in predicting final persuasion ratings than changes. An OLS/Linear Regression model with MAE = 0.6654, RMSE = 0.8727, and $R^2 = 0.7901$ accounted for roughly 79% of the variance in final scores. The triangles follow the line denoting a perfect fit in the first figure, but the prediction models seem to smooth each actual rating, thus what is visible is smoothing error. With an average R^2 of 0.945, the Random Forest regressor outperformed the OLS in modelling non-linear relationships. Its figure shows projected values near to pattern values, especially at higher rating levels (Karinshak et al., 2023). While attitudes toward persuasion shift were easier to predict, the R^2 value of 0.033 was low, limiting its ability to explain attitude shifts. TinyLlama's classification accuracy matches LLaMA-3.2's 0.6389 and macro F1's 0.2599. Although correct, balanced class performance was poor, especially in Negative, Neutral, and Positive shift categories. LLaMA-3.2 was more efficient since it reduced training time while maintaining performance.

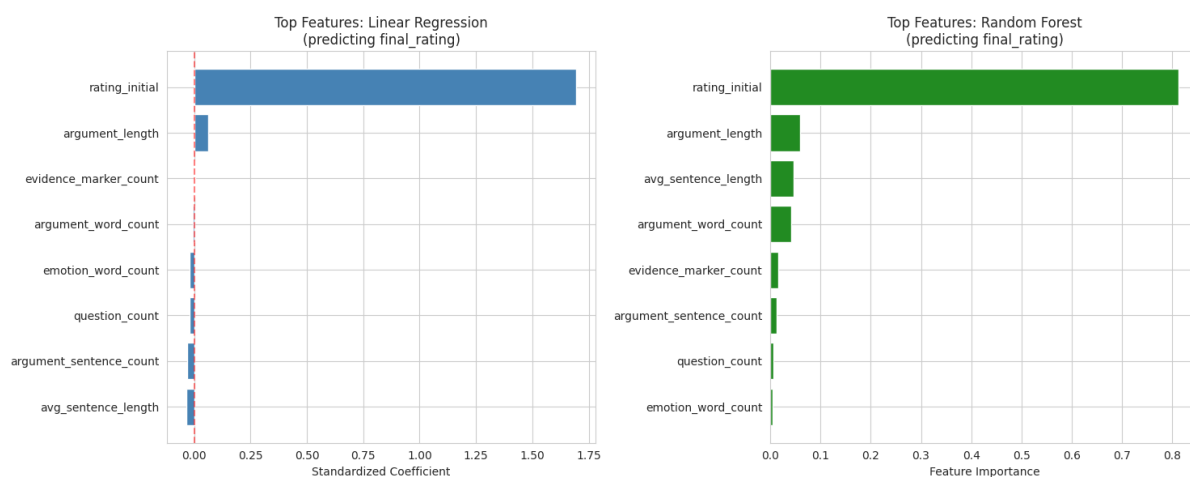


Figure 16: Top predictive features for final rating from linear regression (standardized coefficients) and random forest (feature importance)

6.2 Best Ways to Present the Results

Multiple visualisations of persuasion distribution and causation supported the numerical findings. A histogram of the persuasion shift indicates a strong central tendency with a mean shift of +0.415 and a standard deviation of 0.887, showing that 82.66% of arguments did not affect belief. Boxplots were used to compare the mean shifts for persuasion by human and AI arguments and the mean shifts for specific prompt types within AI arguments. Human arguments had a mean shift of 0.480, AI arguments 0.404, and "Deceptive" 0.470, which was higher than "Infectum" (0.239) and "Warn" (0.454) (Barale, 2021). To demonstrate this strong linear link between the two variables, a scatter plot was generated with most data points near the line "no change" for the initial and final ratings. Additionally, Random Forest model-based feature importance bar charts showed that the first rating is the most essential feature, encompassing 0.812 of feature importance, far more than other textual features like argument length (0.059). To support the visual records with statistical data, correlation heatmaps showed a substantial positive correlation between initial and final scores (0.888) (Ruiz-Arellano et al., 2022).

6.3 What the Results Mean in This Project

The empirical results substantially change computer-based persuasion ideas. According to descriptive data, the average argument only shifts 0.415 on a seven-point belief scale, indicating that most arguments don't modify established beliefs. The OLS regression analysis reveals a significant coefficient of initial rating (1.694; $p < 0.001$), indicating that textual features like argument length and evidence markers had little impact on this weakness (Ruiz-Arellano et al., 2022). The argument a person makes will be influenced by their previous thoughts, even if it is ineffective. The low R^2 of the persuasion shift model (0.033) suggests that additional factors, not just text qualities, contribute to the variance between pre- and post-states. The ANOVA results indicated that prompt types had a substantial impact on final evaluations ($F = 15.31$, $p < 0.001$). Transformer models, with about 0.63 accuracy, show that machine learning can now pick up general persuasion patterns, making the nuanced, three-way shift in human opinions partially solved (Karinshak et al., 2023).

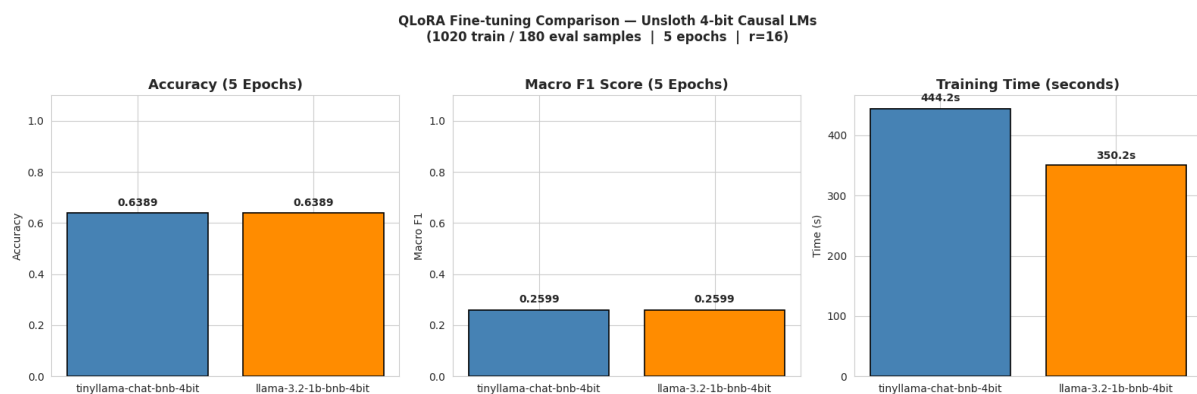


Figure 17: Model Comparison between TinyLlama and LLaMA-3.2 fine-tuned models

6.4 Implications for Real-World Use

Real-world ramifications arise from quantitative AI-human comparisons. The mean final rating for human written arguments (4.877) significantly surpassed AI written arguments (4.564) with P value < 0.0003 , according to an independent t -test. This lower figure suggests that while big language models have improved, AI has yet to fully mimic human persuasive language. The dataset also indicated that the "Deceptive" question (mean shift = 0.470) and "Logical Reasoning" prompt (mean shift = 0.445) had mean shifts that were close to or better than other AI baselines. Fine-tuning a transformer like TinyLlama and DistilBERT to classify persuasion shifts in a text is more than 60% accurate, suggesting automated auditing or text generation (Huang and Wang, 2023). These results cast doubt on AI's general dominance over human persuasion, but with training, coaching, and targeted prompting, AI can methodically frame discourse. To prevent automation-driven, huge manipulation, new and robust policies, digital media principles, and public health measures and capacities are needed.

6.5 Addressing the Research Question

Model	Task Type	Target Variable	MAE	MSE	RMSE	R^2 Score	Overall Interpretation
Linear Regression	Regression	Final rating	0.6654	0.7615	0.8727	0.7901	The Linear Regression model performed well and explained 79.01% of the variance in final rating. It was useful for

							interpretation because it showed the relationship between initial rating, text features, and final persuasion rating.
Random Forest Regressor	Regression	Final rating	0.2976	0.1999	0.4471	0.9449	The Random Forest model achieved the strongest regression performance, with lower error values and a higher R^2 score than Linear Regression. This suggests that non-linear feature interactions improved prediction accuracy.

Model	Task Type	Target Variable	Accuracy	Macro F1	Training Time	Training Loss	Overall Interpretation
unsloth/tinylama-chat-bnb-4bit	Classification	Persuasion shift class: Negative, Neutral, Positive	0.6389	0.2599	483.8 seconds	0.8297	TinyLlama achieved 63.89% accuracy, but the low macro F1 score suggests weaker balanced performance across the three persuasion-shift classes.
unsloth/llama-3.2-1b-bnb-4bit	Classification	Persuasion shift class: Negative, Neutral, Positive	0.6389	0.2599	333.9 seconds	1.3871	Llama 3.2 1B achieved the same accuracy and macro F1 as TinyLlama but trained faster. However, its higher training loss suggests less efficient learning compared with TinyLlama.

The research question is answered by reporting that initial stance, final rating patterns, claim topic and modelling of feature interactions had more significant effects on persuasiveness than text length alone had. This is supported by the results in Figure 6, where argument length is correlated with persuasion shift with a weak correlation, and emotion-word count is also correlated with a weak correlation, while neither sentence count nor evidence-marker count show any significant correlation with the persuasion shift, and the initial and final ratings have a strong correlation with 0.89. This implies that initial positioning of the reader was a crucial determinant in the ultimate persuasiveness. Figures 7 and 8 further show that neither argument frequency nor the most frequently occurring arguments led to greater persuasion (nor did the least frequent arguments). This evidence was stronger in the figures of initially opposed and neutral audience member persuasion shifts when compared to supportive audience member figures such as in Figure 9 and Figure 11, indicating that openness of the audience affected the persuasion result. The analysis of distinctive words in Figure 12 and Figure 13 revealed that there is a difference between high persuasion and low persuasion arguments, but rhetorical feature distributions in Figure 14 did not seem to be entirely distinct, implying that the rhetorical features alone were not sufficient to account for the difference in persuasion between high and low persuasion arguments. Prompt type and AI source had an interaction; Claude 3 Opus typically had higher final ratings (Figure 15). The results of this interpretation are supported by the model results where the Random Forest has the best regression performance at $R^2 = 0.9449$.

CHAPTER 7: ANALYSIS AND DISCUSSION

7.1 Meaning of the Results

The analysis suggests that there is imperative persuasion in this project, though constrained, which, in turn, is a significant outcome. Instead of demonstrating how arguments lead to a complete reversal of earlier views, the findings show that most answers shift by only small amounts after exposure to an argument. This implies that the project is not capturing mere message success or failure, but a more authentic trend of the potential attitude change that can be achieved, but is limited to where the reader begins (Huang and Wang, 2023). The initial rating appears to be the strongest predictor of the final rating in this dataset, while single-surface features such as text length are less important. That trend is important because it indicates that audience preparation has at least as much effect on persuasion as prices.

7.2 Which Model Works Best and Why

Linear regression is most interpretable, whereas Random Forest is best for prediction. The difference is significant. Linear regression estimates a linear relationship by minimising the sum of squared residuals, hence it works when characteristics and outcomes are somewhat direct and additive (Soliman, 2024). This project excels in interpretability because the coefficients show the direction and relative contribution of predictors such as the source, prompt, and initial stance. However, persuasion statistics may not be linear because preexisting beliefs, rhetorical signals, and source labelling might affect an argument's effect. Random Forest works well in this setting because it averages many decision trees to increase predictive performance and decrease overfitting. That makes it better for capturing non-linear correlations and threshold effects in persuasion tasks. For instance, a passionate reader may react differently to the same argument than a neutral reader. A straight-line equation may not show that tendency. Thus, Random Forest's lower error than linear regression in this project may be due to its better data structure (Dehnert and Mongeau, 2022).

7.3 Comparison with the Literature

The project's findings are consistent with the literature discussed above, but more conservative than recent media assertions about AI-persuasion studies. According to Anthropic, model generation boosted persuasiveness and the Claude 3 Opus did not significantly differ from human-written arguments in their benchmark. Lin et al. found pre-registered trials suggesting that AI chats could affect voter opinions more than standard video ads (Karinshak et al., 2023). These works suggest AI can be objectively convincing. The current project appears less brazen, and persuasion relies on previous position and circumstances rather than just the source. That match is that AI-based persuasion must be seen as a communication process driven by cues, context, and human perception, not just a message delivery. It is also consistent with Millet et al. (2023), who discovered that AI-generated outputs were evaluated poorer, demonstrating that source identification can affect evaluations independent of content quality. This underground project does not contradict literature.

7.4 Limitations and Practical Use

Its findings have notable flaws; first, the dataset is limited to short-term opinion changes and does not focus on long-term behavioural changes. It only represents a small part of the field of persuasion; early positions are more predictive than anticipation, which does not consider various other significant dimensions of persuasion, including rhetorical formats

and emotional casing. Its generalizability is limited because the data are confined to an experimental system and do not cover all real-world cases of persuasion (Ruiz-Arellano et al., 2022). Whereas the models can foresee persuasive effects and strategies in certain circumstances, such as education or health communication, there are ethical issues associated with their application, especially in politics, where success in persuasion does not necessarily imply truthfulness.

CHAPTER 8: CONCLUSION

In conclusion, the quantitative analysis of 3,939 records reveals that arguments produced by AI are empirically persuasive, just to a slightly lesser degree (mean shift of 0.415). Initial stance was the strongest predictor of final attitude, with a strong correlation coefficient of 0.888 between ratings of initial and final attitude, and accounting for 79% of the variance in the OLS regression model. Overall, a mean shift of 0.470 was obtained using the AI arguments compared to a human baseline of 0.877, with a p value of 0.0003, whereas more specific AI prompts such as “Deceptive” reasoning showed a competitive mean shift of 0.470. Persuasion, therefore, still continues to be a complex process with much of the persuasion work predicated on the interior of the recipient and his or her opinions, rather than on the naked fact of the writer. The conclusions from these empirical findings are not just theoretical but also have practical significance for health communication and policy messages that should be considered while crafting messages dictated by algorithms. Future studies should build on the use of more advanced linguistic features and test the significance of immediate changes in rating and the subsequent long-term behavioural changes on outcomes.

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APPENDIX